

Single Qubit Benchmarking on a Deuterated Water NMR System

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Abstract

We present an automated simulation-to-NMR pipeline for benchmarking single-qubit control on a benchtop deuterated-water NMR system. The pipeline generates the full 24-element single-qubit Clifford group, simulates Standard Randomized Benchmarking (RB), Interleaved Randomized Benchmarking (IRB), and Deterministic Benchmarking (DB), compiles Clifford operations into hardware-native R_x/R_y pulse sequences, and produces Prospa-compatible experiments for execution on a Magritek Spinsolve spectrometer. Standard RB yielded a depolarizing parameter $p = 0.97614 \pm 0.00447$, corresponding to an average Clifford fidelity $\bar{F}_C = 0.98807 \pm 0.00223$. IRB estimated gate-dependent fidelities: $\bar{F}_S = 0.98902 \pm 0.00413$, $\bar{F}_H = 0.97821 \pm 0.00703$, $\bar{F}_X = 0.97721 \pm 0.00420$, $\bar{F}_{\sqrt{Y}} = 0.98556 \pm 0.00487$, and $\bar{F}_Z = 0.97395 \pm 0.00528$. These values are best interpreted through the physical R_x/R_y pulse decompositions rather than Clifford labels alone. DB was implemented but limited by RF duty-cycle constraints: although independent inversion recovery measured $T_1 = 8.81$ s, safe DB pulse trains were restricted to millisecond-scale windows, preventing reliable extraction of long-time relaxation parameters. The pipeline quantifies average and gate-dependent control fidelity and identifies RF duty cycle as the principal limitation for full deterministic benchmarking.

1 Introduction

Accurate quantum gate characterization is essential for validating quantum hardware and improving control. Small per-gate errors accumulate over circuit depth, so benchmarking protocols must estimate gate performance while separating intrinsic control errors from state-preparation and measurement (SPAM) errors. Randomized Benchmarking (RB) addresses this by fitting the survival probability of random Clifford sequences to a single-exponential decay, yielding an average Clifford error rate that is relatively insensitive to SPAM errors [2]. Interleaved Randomized Benchmarking (IRB) extends this idea by inserting a fixed target gate between random Cliffords, providing a gate-specific fidelity estimate relative to the Standard RB reference.

Nuclear magnetic resonance (NMR) is a useful platform for studying such benchmarking protocols because single-spin rotations can be implemented through calibrated radiofrequency (RF) pulses and measured through ensemble magnetization. However, translating abstract benchmarking circuits into NMR experiments requires careful pulse compilation, phase control, reference normalization, and attention to hardware limits such as RF duty cycle and drift. These practical constraints are especially important for long pulse sequences, where the assumptions of idealized benchmarking protocols may fail.

In this work, we benchmark single-qubit control on a benchtop Magritek Spinsolve 60 MHz spectrometer using a 5% H_2O / 95% D_2O sample. In prior use of this apparatus, gate fidelity was treated as an assumed parameter rather than directly measured. We therefore develop an end-to-end simulation-to-NMR pipeline that generates the 24 single-qubit Clifford gates, simulates RB/IRB/DB sequences, compiles Clifford operations into hardware-native R_x/R_y pulses, and automatically produces Prospa-compatible experiment files.

A key feature of the pipeline is that it makes the physical pulse realization of each Clifford gate explicit. Although Clifford gates are abstractly defined by their action on Pauli operators, they have different physical implementations on the NMR hardware. Some gates require a single $\pi/2$ pulse, others require a full π pulse, and some require composite sequences. In particular, the Z gate is implemented here as

$R_x(\pi/2)R_y(\pi)R_x(-\pi/2)$ rather than as a virtual frame update. Therefore, gate-dependent fidelities must be interpreted through pulse angle, RF exposure, axis calibration, and composite-pulse structure.

We apply Standard RB, IRB, and DB to this system. RB provides the average Clifford fidelity, IRB estimates the fidelities of selected gates such as S , H , X , $\sqrt{Y}$, and Z , and DB tests whether the observed errors can be related to physical parameters such as T_1 , T_2 , $\delta\theta$, and $\delta\phi$ [1]. The RB and IRB experiments provide meaningful average and gate-dependent fidelity estimates, while the DB experiment reveals a practical hardware limitation: dense RF pulse trains are restricted to millisecond-scale windows by duty-cycle constraints, whereas the measured relaxation time is $T_1 = 8.81$ s. Thus, the DB fits are treated as short-time phenomenological fits rather than as a complete Lindblad-model reconstruction.

2 Theory and Simulation

This section develops the theoretical framework for the three benchmarking protocols and presents Python simulation outputs that serve as ground-truth benchmarks against which the NMR results are compared.

2.1 Gate Errors and CPTP Formalism

An ideal quantum gate implements a unitary operation U on a quantum state ρ via the superoperator

$$\mathcal{U}(\rho) = U\rho U^\dagger. \quad (1)$$

In practice, the actual operation is

$$\tilde{\mathcal{U}} = \mathcal{E} \circ \mathcal{U}, \quad (2)$$

where $\mathcal{E}$ absorbs all errors acting on the gate [1]. Gate errors fall into two categories.

Coherent errors preserve state purity but rotate the system to an incorrect state. On the Bloch sphere, they manifest as deviations in the rotation angle ($\delta\theta$) and the rotation axis ($\delta\phi$). Coherent errors add as amplitudes rather than probabilities and are therefore more damaging than incoherent errors at equivalent magnitude.

Incoherent errors degrade state purity and stem from environmental coupling, characterised by the relaxation time T_1 and the coherence time T_2 . They are modelled by the depolarizing channel:

$$\mathcal{E}_{\text{dep}}(\rho) = (1 - q)\rho + \frac{q}{3}(X\rho X + Y\rho Y + Z\rho Z). \quad (3)$$

For a single qubit ($D = 2$), a general completely-positive trace-preserving (CPTP) map has at most $D^4 - D^2 = 12$ independent real parameters. The DB protocol reduces this count to the four physically relevant parameters listed in Table 1 [1].

Parameter	Definition
T_1	Relaxation Time
T_2	Total Coherence Time (including T_ϕ)
$\delta\theta$	Rotation Error
$\delta\phi$	Phase Error

Table 1: Table of DB-Induced Physical Parameters

Incoherent dynamics during gate operation are described by the Lindblad master equation:

$$\mathcal{L}(\rho) = -i[H(t), \rho] + \sum_{\alpha \in \{1, \phi\}} \gamma_\alpha (L_\alpha \rho L_\alpha^\dagger - \frac{1}{2}\{L_\alpha^\dagger L_\alpha, \rho\}), \quad (4)$$

with jump operators $L_1 = |0\rangle\langle 1| = \sigma_-$ and $L_\phi = \sigma_z/\sqrt{2}$. The corresponding rates are $\gamma_1 = 1/T_1$ (relaxation) and $\gamma_\phi = 1/T_\phi$ (pure dephasing), where the total coherence time satisfies $1/T_2 = 1/(2T_1) + 1/T_\phi$.

2.2 Standard Randomized Benchmarking (RB)

2.2.1 Framework

The single-qubit Clifford group $\mathcal{C}_1$ consists of all unitaries C that normalise the Pauli group under conjugation:

$$CPC^\dagger \in \mathcal{P}_1, \quad \forall P \in \mathcal{P}_1. \quad (5)$$

Standard RB assumes gate-independent noise and extracts an average gate infidelity by running K random Clifford sequences for each sequence length n . Each sequence is followed by a classically computed inversion gate, and the average survival probability $P_{\text{avg}}(n)$ is measured over the K sequences. Twirling the error map over the Clifford group reduces it to a depolarizing channel, so the survival probability decays as

$$P_{\text{avg}}(n) = Ap^n + B, \quad (6)$$

where A and B absorb SPAM errors (satisfying $A + B \approx 1$) and p is the depolarizing parameter. The average Clifford gate infidelity is

$$r_C = \frac{D-1}{D}(1-p) \xrightarrow{D=2} \frac{1}{2}(1-p). \quad (7)$$

The uncertainty is propagated from the covariance of the fit: $\sigma(r_C) = \sigma(p)/2$.

2.2.2 Python Simulation

The full 24-element single-qubit Clifford group is generated via breadth-first search (BFS) over the generator set $\{H, S, H^\dagger, S^\dagger\}$, using a phase-invariant canonical matrix key for gate identification.

For each Clifford element, the simulator assigns a human-readable label. The seven named gates ($I, X, Y, Z, H, S, S^\dagger$) retain their standard labels; the remaining seventeen are labelled by their Pauli conjugation signature, formed by concatenating the images of $\{X, Y, Z\}$ under $UPU^\dagger$.

The simulator completes one RB sequence of length n as follows:

1. The qubit is initialised to $\rho = |0\rangle\langle 0|$.
2. At each step, a random Clifford gate G is drawn uniformly from the gate set.
3. G is applied as $\rho \rightarrow G\rho G^\dagger$, immediately followed by a depolarizing channel with $q \sim \mathcal{N}(\mu, \sigma^2)$ (clipped to $[0, \infty)$), applied via Eq. (3).
4. After n such steps, the cumulative unitary $U_{\text{cum}} = G_n \cdots G_1$ is computed and the inversion gate C_{inv} satisfying $C_{\text{inv}}U_{\text{cum}} = \alpha I$ is located in the Clifford set.
5. The survival probability $s = \text{Tr}(|0\rangle\langle 0| \rho)$ is clipped to $[0, 1]$ and binomially sampled with $N_{\text{shots}} = 1024$.
6. This is averaged over $K = 32$ independent runs per sequence length to obtain $P_{\text{avg}}(n)$ and its standard error.
7. The averaged data are fit to Eq. (6) using `scipy.optimize.curve_fit` with initial guess $(A, p, B) = (0.5, 0.9, 0.5)$ and bounds $A, p, B \in [0, 1]$.
8. The extracted p is converted to the average gate infidelity via Eq. (7).
9. All gate sequences are exported to a CSV file for later use by the NMR compiler.

The simulation parameters are listed in Table 2.

Parameter	Value	Description
N_{max}	128	Maximum sequence length
K	32	Sequences per length
N_{shots}	1024	Measurement shots per run
μ	0.01789	Mean depolarizing noise
σ	0.00179	Noise standard deviation
SEED	42	RNG seed for reproducibility

Table 2: Global parameters for the Standard RB simulation.

Since this simulation uses a gate-independent depolarizing noise model, it should be interpreted as a baseline single-exponential RB benchmark rather than as a pulse-resolved model of individual R_x/R_y hardware errors. The resulting decay curve is shown in Figure 1.

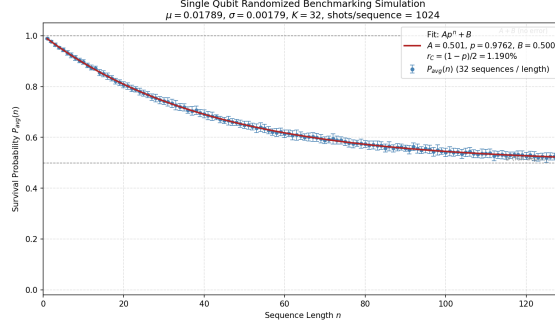


Figure 1: Simulated Standard RB decay with fitted A , p , B , and r_C values.

2.3 Interleaved Randomized Benchmarking (IRB)

2.3.1 Framework

IRB extracts the infidelity of a specific target Clifford gate $\mathcal{G}$ [1] by interleaving it between every pair of random Clifford gates:

$$\mathcal{U}_{n,\text{IRB}} = C_n \cdot \mathcal{G} \cdot C_{n-1} \cdot \mathcal{G} \cdot C_{n-2} \cdots \mathcal{G} \cdot C_1. \quad (8)$$

IRB produces a second decay parameter p_{inter} , from which the gate-specific infidelity is

$$r_{\mathcal{G}} = \frac{D-1}{D} \left(1 - \frac{p_{\text{inter}}}{p} \right) \xrightarrow{D=2} \frac{1}{2} \left(1 - \frac{p_{\text{inter}}}{p} \right). \quad (9)$$

The propagated uncertainty is $\sigma(r_{\mathcal{G}}) = \sigma(p_{\text{inter}})/(2p)$.

Because coherent errors in the target gate may constructively or destructively interfere with errors from the random Cliffords (potentially yielding $p_{\text{inter}} > p$) [1], the true gate error lies within the interval $[r_{\mathcal{G}} - E, r_{\mathcal{G}} + E]$, where

$$E = \min \left\{ \frac{(D-1)[|p - p_{\text{inter}}/p| + (1-p)]}{Dp}, \frac{2(D^2-1)(1-p)}{pD^2} + \frac{4\sqrt{1-p}\sqrt{D^2-1}}{p} \right\}. \quad (10)$$

This systematic uncertainty of IRB motivates DB as a complementary protocol for characterising coherent gate errors.

Tripathi *et al.* [1] provide a hierarchy of pulse types from which each Clifford gate is assigned a noise model $(\mu_{\mathcal{G}}, \sigma_{\mathcal{G}})$ based on its physical implementation:

Gate	Physical Basis
I, Z	Virtual gate (minimal noise)
$S, S^\dagger$	$\pi/2$ pulses (slight noise)
H	$\pi/2 + \pi/2$ composite (moderate noise)
X, Y	π pulses (highest noise)
Remaining Cliffords	Multi-pulse composites

Table 3: Clifford gates categorised by physical pulse basis and expected noise level.

Virtual gates contribute negligible noise, $\pi/2$ pulses accumulate slight depolarizing error, and π pulses accumulate the most.

2.3.2 Python Simulation

Unlike the Standard RB simulation, the IRB simulation incorporates the hardware-native R_x/R_y pulse decomposition of each target Clifford gate. Since the NMR system implements the Z gate as the composite sequence $R_x(\pi/2) R_y(\pi) R_x(-\pi/2)$ rather than as a virtual frame update, the noise of each target gate is calculated by combining the depolarizing channels of its constituent physical pulses via the shrinkage product:

$$q_{\text{eff}} = \frac{3}{4} \left[1 - \prod_i \left(1 - \frac{4q_i}{3} \right) \right]. \quad (11)$$

The simulation derives gate-specific noise from two primitive pulse parameters: $q_{\pi/2}$ for a single $R_x/R_y(\pi/2)$ pulse and q_π for a single $R_x/R_y(\pi)$ pulse.

The two q values are extracted from the NMR IRB measurements ($p_{\text{inter},X}$, $p_{\text{inter},S}$) by inverting the IRB expression. The X gate anchors $q_\pi = \frac{3}{4}(1 - p_{\text{inter},X}/p)$, and the S gate anchors $q_{\pi/2}$ via $\frac{3}{4}[1 - (1 - \frac{4q_{\pi/2}}{3})^3] = \frac{3}{4}(1 - p_{\text{inter},S}/p)$. From these values, the global RB noise parameter $\mu = \frac{3}{4}(1 - p_{\text{NMR}})$ is recovered, and all remaining Clifford gate noises are derived from the two primitive q values.

The five target gates and their noise models are listed in Table 4.

Gate	R_x, R_y Decomposition	Noise Model
X	$1x\pi$	$(q_\pi, 0.1q_\pi)$
H	$\pi + \pi/2$	$(\text{shrink}(q_\pi, q_{\pi/2}), 0.00300)$
S	$3x\pi/2$	$(\text{shrink}(q_{\pi/2}, q_{\pi/2}, q_{\pi/2}), 0.00150)$
Z	$\pi/2 + \pi + \pi/2$	$(\text{shrink}(q_{\pi/2}, q_\pi, q_{\pi/2}), 0.00100)$
$\sqrt{Y}$	$1x\pi/2$	$(\text{shrink}(q_{\pi/2}), 0.00070)$

Table 4: IRB target gates with pulse-composition-aware noise models ($K_{\text{IRB}} = 64$; lower confidence bound clamped to zero and flagged when $r_G \leq E$).

The resulting simulated IRB decay curves are shown in Figure 2, overlaid on the Standard RB baseline. These results motivate interpreting experimental IRB fidelities through the physical R_x/R_y decompositions in Table 6, rather than through a simple rule that more pulses always imply lower fidelity.

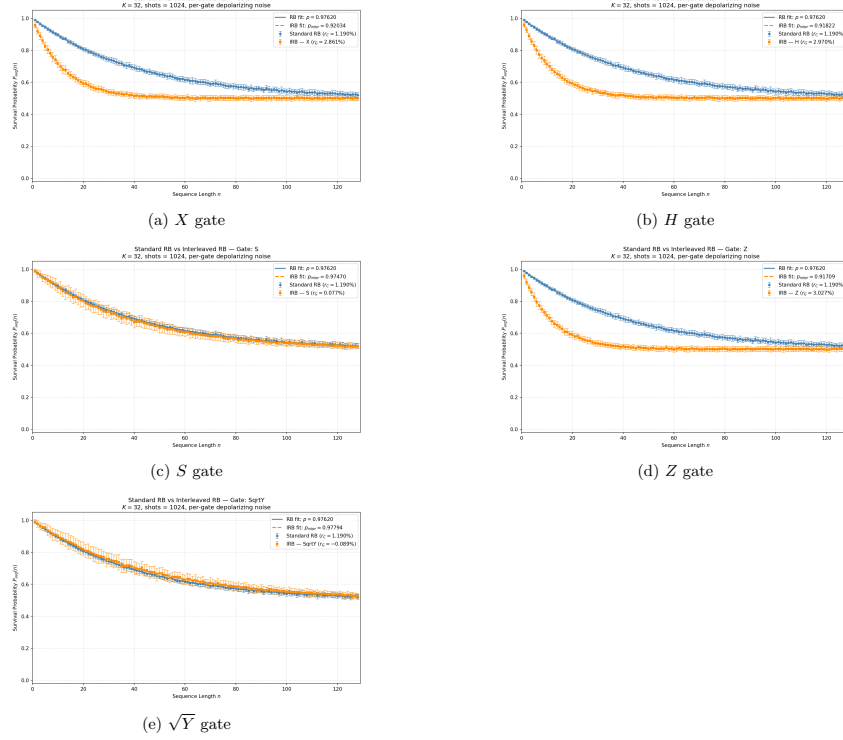


Figure 2: Simulated IRB decay curves for each target gate, overlaid on the Standard RB reference.

2.4 Deterministic Benchmarking (DB)

2.4.1 Protocol

DB extracts four key parameters $\{T_1, T_2, \delta\theta, \delta\phi\}$ from four fixed gate-pair sequences, eliminating the need for random-circuit averaging [1]. The gate pairs are drawn from $\{XX, X\bar{X}, YY, \bar{Y}Y, Y\bar{Y}\}$, where a bar denotes negative error signs (e.g., $\bar{X} \equiv R_x(-\pi - \delta\theta) \cdot R_z(-\delta\phi)$). After n repetitions, the fidelity evolves as [1]

$$F(t_n) = \frac{1}{2}(1 + a) + \frac{1}{2}(1 - a)e^{-t_n/T_D} \cos(2\omega t_n), \quad (12)$$

where a parameterises the long-time saturation fidelity ($t \rightarrow \infty$), T_D is the incoherent decay timescale, and ω is the coherent oscillation frequency.

2.4.2 Four-Experiment Framework

The four experiments and their target parameters are:

1. $\{\text{Free}; |1\rangle\}$: Free evolution (no pulses) extracts T_1 . Eq. (12) reduces to $\frac{1}{2}(1 + a) + \frac{1}{2}(1 - a)e^{-t/T_1}$.
2. $\{XX; |+\rangle\}$: The XX sequence is insensitive to coherent errors, so the fitted decay gives $T_D = T_2$.
3. $\{YY; |+\rangle\}$: Fidelity oscillates at $\omega \approx \delta\theta/(2t_g)$, yielding $\delta\theta = 2\omega t_g$.
4. $\{X\bar{X}; |+\rangle\}$: Fidelity oscillates at $\omega \approx \delta\phi/(2t_g)$, yielding $\delta\phi = 2\omega t_g$.

2.4.3 Python Simulation

The simulation introduces noisy gates as $U_{\text{noisy}} = R_\alpha(\pm\pi + \delta\theta) \cdot R_z(\pm\delta\phi)$ and implements five gate pairs: $XX, X\bar{X}, YY, \bar{Y}Y$, and $Y\bar{Y}$.

The Lindblad superoperator

$$\mathcal{L} = \gamma_1 \mathcal{D}[L_1] + \gamma_\phi \mathcal{D}[L_\phi/\sqrt{2}], \quad (13)$$

with L_1 , L_ϕ , γ_1 , γ_ϕ as defined in Section 3.1, is exponentiated using `scipy.linalg.expm` to create the open-system propagator $E_{\text{open}}(t_g) = e^{\mathcal{L}t_g}$, with the density matrix vectorised in column-major order. One application of a gate pair (U_1, U_2) is $E_{\text{pair}} = E_{\text{gate}}(U_2) \circ E_{\text{gate}}(U_1)$, where $E_{\text{gate}}(U) = E_{\text{open}}(t_g) \circ U$.

The simulator fits Eq. (12) using `scipy.optimize.curve_fit`. For experiments 1 and 2, $\omega = 0$ and (a, T_D) are free parameters. For experiments 3 and 4, (a, T_D, ω) are free, with ω initialised from a fast Fourier transform of the oscillations. The four physical parameters are then recovered using the relations in Section 3.3.1.

The simulation uses DB parameters matching Tripathi *et al.* [1]: $\{T_1 = 23.36 \mu\text{s}, T_2 = 44.12 \mu\text{s}, \delta\theta = 0.398^\circ, \delta\phi = 0.426^\circ, t_g = 88 \text{ ns}\}$. The resulting output is shown in Figure 3.

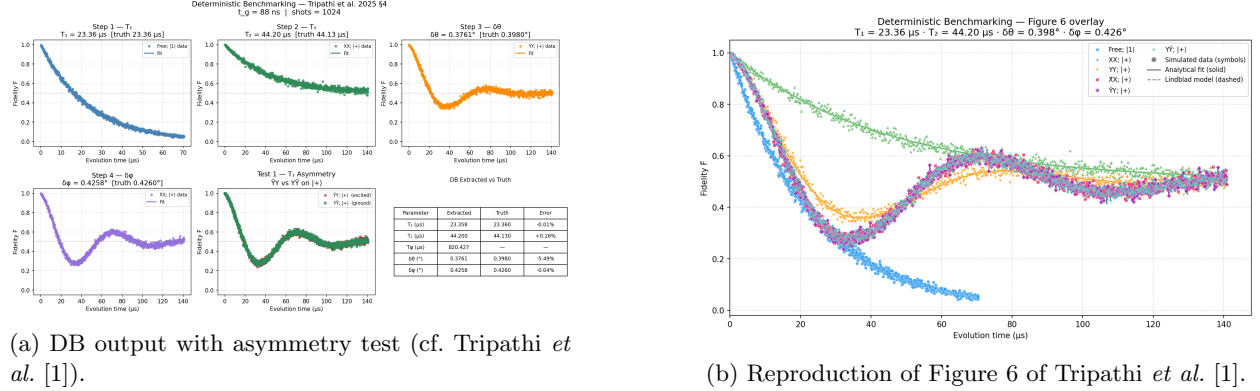


Figure 3: Simulated DB fidelity decay curves.

3 Experiments

3.1 Apparatus and Experimental Parameters

All NMR experiments were performed on a benchtop Magritek Spinsolve 60 MHz spectrometer (Magritek GmbH), operating at a ^1H Larmor frequency of 60 MHz. The sample is a mixture of 5% water and 95% heavy water (D_2O) by volume. The solution is dilute enough that interactions between different molecules can be neglected. All experiments were performed at room temperature, and the sample was prepared in a 5 mm outer-diameter NMR tube filled above the instrument’s minimum sample volume.

To ensure consistency, a standardized set of acquisition parameters was maintained across all calibration, T_1 recovery, and benchmarking sequences (RB, IRB, DB), except where explicitly swept as an experimental variable. These global hardware parameters are summarized in Table 5.

Parameter	Value
Centre Frequency	4.72 ppm
Pulse Amplitude	0 dB
Receiver Channel	^1H
Receiver Gain	25
Number of points	4096
Dwell Time	1000 μs
Repetition Time (T_{rep})	30,000 ms
Receiver Phase	177°
Number of scans	1

Table 5: Standard acquisition parameters used throughout all NMR benchmarking experiments.

Crucially, empirical testing during the campaign revealed stringent hardware duty-cycle constraints. When subjecting the apparatus to dense, continuous RF driving, we observed that pulse sequences extending

beyond approximately 260 discrete pulses induced significant physical drift and spectrometer stress, causing the measured signal to unrecoverably degrade. This duty-cycle ceiling restricts the operational capability of the device to the short-time (millisecond) measurement regime. As a direct consequence, we were physically unable to perform full-length deterministic benchmarking protocols, which require interrogating the system over multiple seconds to accurately extract long-time relaxation dynamics. The implications of this short-time limitation will be discussed further in Section 5.4.

3.2 Pulse Duration Calibration ($\pi/2$)

The amplitude and duration of a radiofrequency (RF) pulse determine the rotation angle of the net magnetization vector about the effective magnetic field in the rotating frame. A $\pi/2$ pulse rotates the equilibrium magnetization from the z -axis into the transverse plane, producing maximum signal. By varying the pulse duration and measuring the corresponding signal intensity, one observes a sine-like dependence, allowing calibration of the optimal $\pi/2$ pulse length.

In this experiment, the $\pi/2$ pulse length was calibrated by first performing a rough sweep from 1–20 μs to locate the approximate $\pi/2$ pulse duration, followed by a precise sweep between 10.5–14.5 μs . As shown in Figure 4, the signal intensity showed a sine-like dependence on pulse duration with a maximum at 13.0 μs corresponding to the $\pi/2$ pulse. This measured value serves as the foundational pulse length $t_{\pi/2}$ utilized in all subsequent benchmarking sequences.

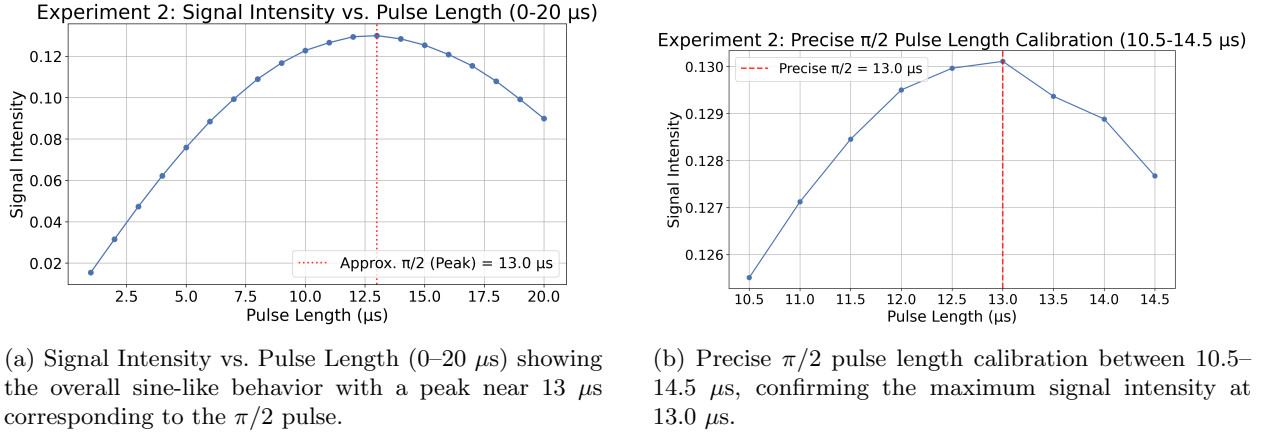


Figure 4: Pulse Duration Calibration ($\pi/2$). The first sweep (a) identifies the approximate $\pi/2$ pulse length, while (b) refines the calibration to identify a precise optimal duration of 13.0 μs .

3.3 T_1 Measurement: Inversion Recovery

The longitudinal (spin-lattice) relaxation time T_1 characterises how quickly the net magnetisation recovers along B_0 after a perturbation. Knowledge of T_1 is essential for the benchmarking pipeline: it sets the minimum repetition time required between successive sequences ($T_{\text{rep}} \geq 5T_1$), enters directly into the noise model used by the DB simulator, and provides an upper bound on the coherence time available for long benchmarking sequences.

Using the inversion recovery method, an initial π pulse inverts the equilibrium magnetisation M_0 to $-M_0$. After a variable delay τ , a $\pi/2$ readout pulse converts the recovered longitudinal magnetisation into the transverse plane for detection. The signal amplitude evolves as $M_z(\tau) = M_0(1 - 2e^{-\tau/T_1})$. Fitting experimental amplitudes across varying delay times yields the relaxation rate.

As shown in Figure 5, the fit yields $T_1 = 8.81\text{ s}$ for the ^1H spins in the 5%/95% $\text{H}_2\text{O}/\text{D}_2\text{O}$ sample. This long relaxation time dictates a repetition time of at least $5T_1 \approx 44\text{ s}$ to ensure full magnetisation recovery between scans. In the context of randomised benchmarking, it also means that the average physical RF gate time of $\sim 28\text{ }\mu\text{s}$ is over five orders of magnitude shorter than T_1 , so longitudinal relaxation contributes negligibly to per-gate error on the time-scale of a single Clifford gate.

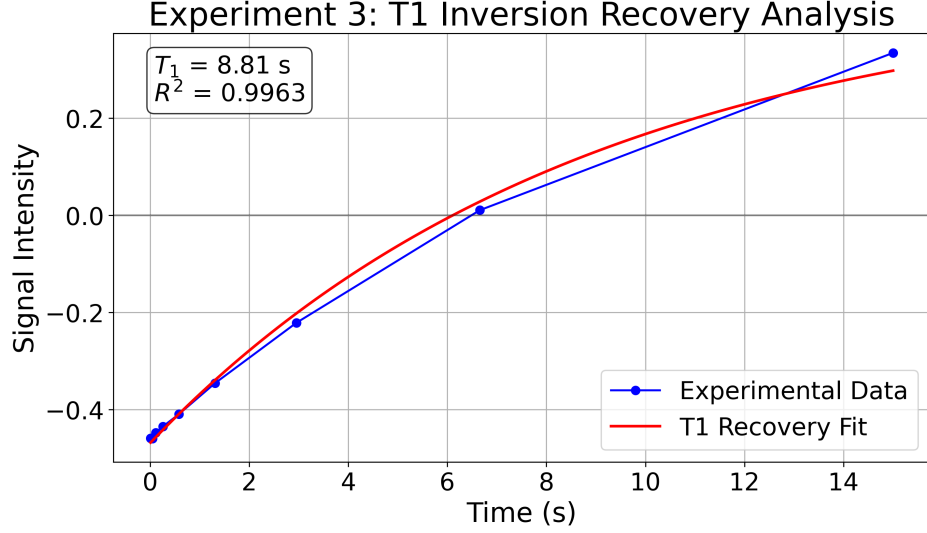


Figure 5: T_1 inversion-recovery analysis for ^1H water. Signal intensities (blue) are fitted to $M_z(\tau) = M_0(1 - 2e^{-\tau/T_1})$ (red). The best-fit yields $T_1 = 8.81 \text{ s}$ with $R^2 = 0.9963$, indicating excellent agreement between experiment and theory.

3.4 Clifford Gates to NMR Pulse Decomposition

The 24 single-qubit Clifford gates correspond to the rotational symmetries of the octahedron—the unitaries U (up to global phase) satisfying $U\sigma_\alpha U^\dagger \in \{\pm\sigma_x, \pm\sigma_y, \pm\sigma_z\}$.

Each gate is decomposed into hardware-native R_x/R_y pulses only. Virtual R_z rotations are eliminated using the identity

$$R_z(\theta) = R_x(\frac{\pi}{2}) \cdot R_y(\theta) \cdot R_x(-\frac{\pi}{2}), \quad (14)$$

where $R_\alpha(\theta) = e^{-i\theta\sigma_\alpha/2}$.

The full list of single-qubit Clifford gates, their Pauli conjugation mappings, and their R_x/R_y -only pulse decompositions is summarized in Table 6.

#	Name	$X \rightarrow$	$Y \rightarrow$	$Z \rightarrow$	R_x/R_y only	Pulses
0	I	$+X$	$+Y$	$+Z$	I	0
1	X	$+X$	$-Y$	$-Z$	$R_x(\pi)$	1
2	Y	$-X$	$+Y$	$-Z$	$R_y(\pi)$	1
3	Z	$-X$	$-Y$	$+Z$	$R_x(\frac{\pi}{2})R_y(\pi)R_x(-\frac{\pi}{2})$	3
4	$\sqrt{X}$	$+X$	$+Z$	$-Y$	$R_x(\frac{\pi}{2})$	1
5	$\sqrt{X}^\dagger$	$+X$	$-Z$	$+Y$	$R_x(-\frac{\pi}{2})$	1
6	$\sqrt{Y}$	$-Z$	$+Y$	$+X$	$R_y(\frac{\pi}{2})$	1
7	$\sqrt{Y}^\dagger$	$+Z$	$+Y$	$-X$	$R_y(-\frac{\pi}{2})$	1
8	S	$+Y$	$-X$	$+Z$	$R_x(\frac{\pi}{2})R_y(\frac{\pi}{2})R_x(-\frac{\pi}{2})$	3
9	$S^\dagger$	$-Y$	$+X$	$+Z$	$R_x(\frac{\pi}{2})R_y(-\frac{\pi}{2})R_x(-\frac{\pi}{2})$	3
10	H	$+Z$	$-Y$	$+X$	$R_x(\pi)R_y(\frac{\pi}{2})$	2
11	$H_{\bar{x}\bar{z}}$	$-Z$	$-Y$	$-X$	$R_x(\pi)R_y(-\frac{\pi}{2})$	2
12	H_{yz}	$-X$	$+Z$	$+Y$	$R_y(\pi)R_x(-\frac{\pi}{2})$	2
13	$H_{\bar{y}\bar{z}}$	$-X$	$-Z$	$-Y$	$R_y(\pi)R_x(\frac{\pi}{2})$	2
14	H_{xy}	$+Y$	$+X$	$-Z$	$R_x(\frac{\pi}{2})R_y(\frac{\pi}{2})R_x(\frac{\pi}{2})$	3
15	$H_{\bar{x}\bar{y}}$	$-Y$	$-X$	$-Z$	$R_x(\frac{\pi}{2})R_y(-\frac{\pi}{2})R_x(\frac{\pi}{2})$	3
16	V_1	$+Y$	$+Z$	$+X$	$R_x(\frac{\pi}{2})R_y(\frac{\pi}{2})$	2
17	$V_1^\dagger$	$+Z$	$+X$	$+Y$	$R_y(-\frac{\pi}{2})R_x(-\frac{\pi}{2})$	2
18	V_2	$-Z$	$+X$	$-Y$	$R_y(\frac{\pi}{2})R_x(\frac{\pi}{2})$	2
19	$V_2^\dagger$	$+Y$	$-Z$	$-X$	$R_x(-\frac{\pi}{2})R_y(-\frac{\pi}{2})$	2
20	V_3	$-Z$	$-X$	$+Y$	$R_y(\frac{\pi}{2})R_x(-\frac{\pi}{2})$	2
21	$V_3^\dagger$	$-Y$	$+Z$	$-X$	$R_x(\frac{\pi}{2})R_y(-\frac{\pi}{2})$	2
22	V_4	$+Z$	$-X$	$-Y$	$R_y(-\frac{\pi}{2})R_x(\frac{\pi}{2})$	2
23	$V_4^\dagger$	$-Y$	$-Z$	$+X$	$R_x(-\frac{\pi}{2})R_y(\frac{\pi}{2})$	2

Table 6: Single-Qubit Clifford Group Gates and their R_x/R_y -only NMR Pulse Decompositions

The distribution of RF pulse counts and their underlying geometric classes are highly relevant to characterising gate error rates because each physical pulse introduces both systematic calibration error and incoherent decoherence. Across the single-qubit Clifford group, the decomposition yields one trivial gate requiring zero pulses (the identity), six gates requiring a single pulse ($\pm\pi/2$ and π rotations about the coordinate x - and y -axes), twelve gates requiring two pulses, and five gates requiring three pulses.

Geometrically, these 24 gates partition into five classes corresponding to the rotational symmetries of the octahedron: the identity class (1 gate), π -rotations about coordinate axes (3 gates), $\pm\pi/2$ -rotations about coordinate axes (6 gates), π -rotations about face-diagonal axes (6 gates), and $\pm2\pi/3$ -rotations about body-diagonal axes (8 gates). The average number of pulses required to implement an arbitrary Clifford gate in this decomposition is 1.875.

Given the experimentally calibrated 90° ($\pi/2$) pulse length of $t_{\pi/2} = 13 \mu\text{s}$, the average physical RF pulse length is $28.383 \mu\text{s}$ per Clifford gate.

3.5 Automated NMR Sequence Compilation and Batch Execution

Running a full benchmarking campaign (whether RB, IRB, or DB) on the Spinsolve requires hundreds of distinct pulse programs to be compiled into the Prospa-native experiment format and executed in an unattended batch. We developed an end-to-end Python pipeline that automates every step from the simulated sequence output through to instrument execution. The pipeline operates in four stages: (i) sequence generation, (ii) pulse-level decomposition, (iii) Spinsolve experiment compilation, and (iv) orchestrator macro generation.

3.5.1 Benchmarking Sequence Generation

The simulator generates a CSV file in which each row represents a single benchmarking run. The columns encode the sequence length m , the run index k , the simulated survival probability, the required final readout/inversion gate, and the m sampled sequence elements (e.g., random Clifford labels for RB, or repeated target gates for DB). For an RB campaign, the builder script selects a set of geometrically spaced sequence lengths up to a specified maximum $m_{\max}$ and generates K independent random sequences for each retained length. Conversely, for a DB campaign, sequence lengths are simply chosen as multiples of 5 up to the maximum length. The execution order within each k -block is randomized (using a fixed seed for reproducibility) to decorrelate systematic drifts from the sequence-length variable, a design choice motivated by the multi-hour duration of an overnight campaign.

3.5.2 Pulse-Level Decomposition

A converter script reads the generated CSV and replaces each gate label with its hardware-native R_x/R_y -only pulse decomposition, as listed in Table 6. The converter maintains a dictionary of all relevant gates, with each entry storing the ordered list of `Pulse(axis, angle_rad)` objects that constitute the R_x/R_y -only factorization. For each experiment row, the full gate sequence—including any inversion or readout gate appended at the end—is compiled into a flat list of axis-angle pairs by iterating over the gate labels and emitting the constituent pulses in *reversed* order (rightmost factor first), so that the serialized pulse train preserves the correct matrix product when executed sequentially on the spectrometer. The output is written to a TSV file in which the `rxry_sequence` column encodes the flattened pulse train as a pipe-delimited string of `axis:angle_deg` tokens.

Each compiled pulse sequence is additionally validated algebraically (where applicable) to ensure that the serialized operations match the intended target physical state.

3.5.3 Spinsolve Experiment Compilation

The core of the pipeline is a custom NMR compiler module that transforms each flat pulse sequence into a complete Spinsolve experiment directory. The Magritek Spinsolve spectrometer uses the Prospa scripting environment, in which each experiment is defined by a self-contained directory containing four files:

1. A **pulse-program macro** (`.pp.mac`) that contains the low-level pulse sequence—the ordered list of `pulse(channel, amplitude, phase, duration, frequency), delay(...)`, and `acquire(...)` instructions.
2. A **wrapper macro** (`.mac`) that registers the experiment with Prospa’s runtime, defines the pulse-program parameter list (`pp_list`), and sets the phase-cycling table.
3. An **interface macro** (`_interface.mac`) that specifies the GUI controls for user-adjustable parameters such as centre frequency, pulse amplitude, and repetition time.
4. A **default parameter file** (`Default.par`) that provides the initial numeric values for all experiment parameters (pulse lengths, frequencies, gains, etc.).

The compiler is structured around a *scaffold* abstraction: each detection nucleus (^1H or ^{13}C) defines a scaffold that specifies the canonical template files, the mapping from symbolic parameter names to hardware channels (e.g., `a90H` for the ^1H 90° amplitude, `d90H` for its duration), and the default readout block (acquisition pulse followed by a pre-acquisition delay and an `acquire` call). To generate a pulse-program file, the compiler loads the scaffold’s template, locates the placeholder region delimited by the markers `# Your pulse program goes here` and `lst = endpp(0)`, and replaces it with the rendered body operations followed by the readout block. The `relationships` block in the template defines derived parameters such as $d_{180} = 2 \times d_{90}$, the acquisition delay, and the transmit frequencies, ensuring that changes to the 90° pulse length propagate correctly throughout the experiment.

For each sequence row, the compiler accepts a `ProgramSpec` object containing the experiment name and the list of body pulses expressed as typed `RotationPulse(axis, angle_deg, target)` objects. The lowering pass converts each rotation into a concrete `PulseOp` with resolved channel number, amplitude

symbol, phase register (p1 for $+x$, p2 for $+y$, p3 for $-x$, p4 for $-y$), duration symbol (or numeric value for non-standard angles), and transmit frequency. Inter-pulse delays of $0.5\ \mu\text{s}$ are inserted between consecutive pulses to satisfy the spectrometer’s minimum switching time. The four output files are written to a directory named after the experiment (e.g., RB1H_ROW0042/).

3.5.4 Batch Orchestration

The final stage generates a Prospa orchestrator macro that sequentially executes all compiled experiments in a single unattended session. The build script assembles a manifest TSV that assigns a unique manifest index, experiment name, sequence metadata, and a human-readable run label to every job in the campaign—including both the benchmarking sequences and any appended FID reference acquisitions used for signal-amplitude normalization. The orchestrator macro is emitted at the workspace root and performs the following operations for each manifest entry:

1. Checks whether the experiment’s compiled DSP binary (.p file) exists; if not, it invokes the Prospa on-demand compiler (`CompilerBackDoor`) to generate it.
2. Calls `RunExpt(exptName, [parameters])` with overrides for the number of scans, repetition time, calibrated pulse length, and a unique `expName` label that embeds the batch label, sequence length, and run index.
3. Logs structured status lines to a text file after each completed or aborted job, enabling progress monitoring from outside Prospa.

4 Results and Analysis

4.1 Signal Processing and Fidelity Computation

The raw backend observable acquired from the NMR spectrometer is the signed, fixed-phase proton-spectrum contrast, denoted as z . Peak detection is initially performed using the magnitude spectrum, but the final integration employs the phased real spectrum, which preserves the sign of the signal. The integrated peak area is then explicitly normalized by dividing it by the mean signal of the $+Z$ FID references (the free induction decay representing the prepared equilibrium state). This division yields the fractional signal $z \in [-1, 1]$.

To map the normalized NMR contrast $z \in [-1, 1]$ to a probability-like signal $P \in [0, 1]$ that connects directly to the standard benchmarking formalism, we apply the affine transformation:

$$P = \frac{1 + z}{2} \quad (15)$$

The sequence-length dependent survival probability is then modeled by the exponential decay function:

$$P_{\text{avg}}(m) = Ap^m + B \quad (16)$$

Because the transformation is affine, the fitted depolarizing parameter p remains invariant under this mapping. For a single-qubit system ($D = 2$), the average Clifford gate fidelity $\bar{F}_C$ and the specific gate fidelity $\bar{F}_G$ are derived from the fitted decay constants p and p_{inter} as:

$$\bar{F}_C = \frac{1 + p}{2}, \quad \bar{F}_G = \frac{1 + p_{\text{inter}}/p}{2} \quad (17)$$

Uncertainties are reported to 1σ , determined from the nonlinear least-squares fit for p and linearly propagated for the derived quantities.

4.2 Standard Randomized Benchmarking (RB)

The Standard RB protocol was executed to establish the baseline average Clifford gate fidelity of the compiled single-qubit Clifford ensemble. The campaign utilized the appended Free Induction Decay (FID) references at the end of the run as a common normalization scale, ensuring consistency across all sparse sequence lengths. The measured reference decay parameter at the signal peak (± 0.05 ppm) was found to be $p = 0.97614 \pm 0.00447$. Using the relationship defined above, the corresponding average Clifford gate fidelity was computed as:

$$\bar{F}_C = 0.98807 \pm 0.00223 \quad (18)$$

This baseline p acts as the reference depolarization parameter for the subsequent IRB analysis. Since Standard RB averages over the full Clifford ensemble, this value should be interpreted as an effective average fidelity over the R_x/R_y -compiled Clifford gates, rather than as the fidelity of any individual physical pulse. The observed decay is consistent with a single-exponential RB model, although the use of appended FID references assumes that the reference amplitude remains sufficiently stable across the acquisition block.

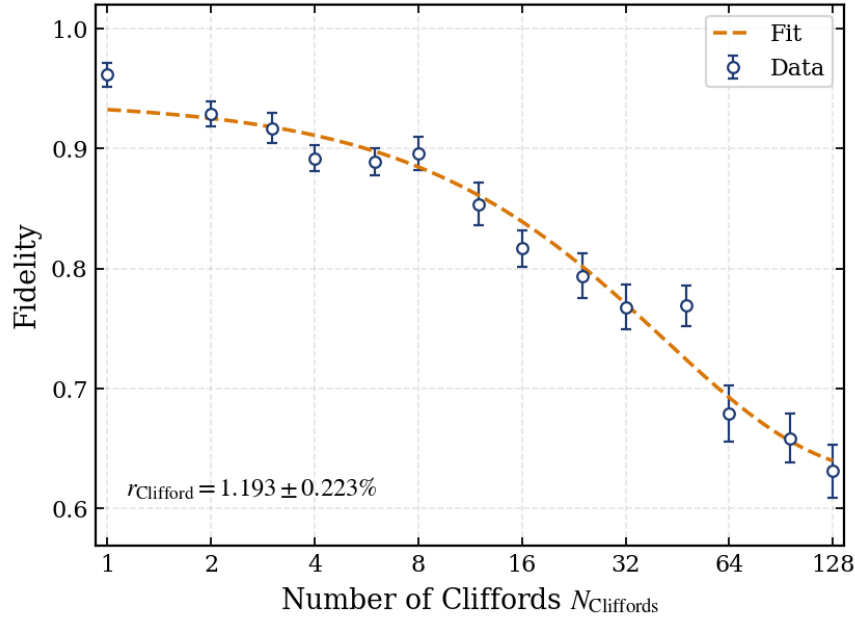


Figure 6: Standard RB decay curve derived from NMR experiment runs. The data are fit to the single-exponential RB model $P_{\text{avg}}(m) = Ap^m + B$.

4.3 Interleaved Randomized Benchmarking (IRB)

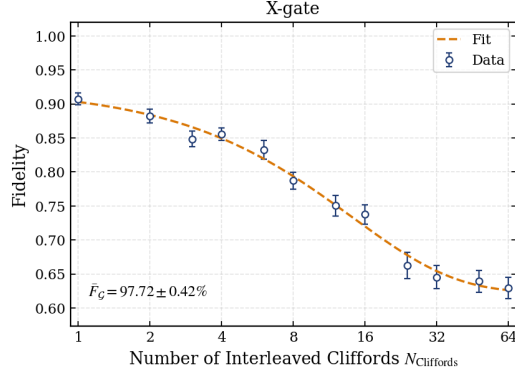
Interleaved RB experiments were performed to extract the error contributions of five specific gates: S , H , X , $\sqrt{Y}$, and Z . For each target gate, 768 IRB runs and 64 FID references were acquired. The FID reference rows were collected in two independent blocks (`first32` and `second32`), and each block was phased and normalized against its own local $+Z$ reference set to mitigate the impact of overnight hardware drift.

The extracted interleaved decay parameters p_{inter} and the calculated individual gate fidelities $\bar{F}_G$ are summarized below:

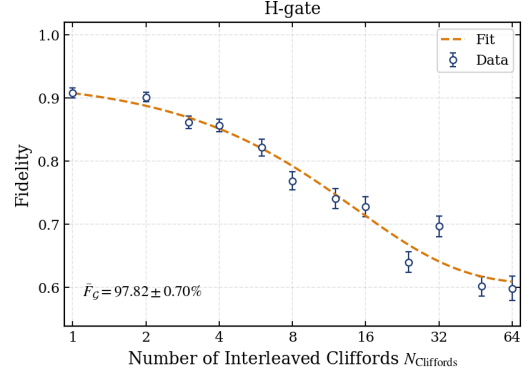
- **S gate:** $p_{\text{inter}} = 0.95470 \pm 0.00678 \implies \bar{F}_G = 0.98902 \pm 0.00413$
- **H gate:** $p_{\text{inter}} = 0.93360 \pm 0.01305 \implies \bar{F}_G = 0.97821 \pm 0.00703$
- **X gate:** $p_{\text{inter}} = 0.93166 \pm 0.00701 \implies \bar{F}_G = 0.97721 \pm 0.00420$
- **$\sqrt{Y}$ gate:** $p_{\text{inter}} = 0.94796 \pm 0.00846 \implies \bar{F}_G = 0.98556 \pm 0.00487$

- **Z gate:** $p_{\text{inter}} = 0.92529 \pm 0.00940 \implies \bar{F}_G = 0.97395 \pm 0.00528$

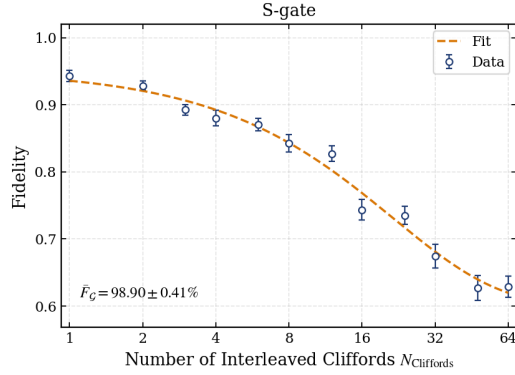
The observed IRB ordering is better explained by the hardware-native R_x/R_y pulse decompositions in Table 6 than by pulse count alone. In particular, the S gate is implemented as $R_x(\pi/2)R_y(\pi/2)R_x(-\pi/2)$ and gives the highest extracted fidelity, even though it contains three physical pulses. This suggests that the relevant error scale is not simply the number of pulses, but the combination of pulse angle, total RF exposure, axis calibration, and possible coherent-error cancellation within composite decompositions. Gates containing a full π pulse, such as X , H , and the synthesized Z gate, are expected to be more sensitive to amplitude miscalibration and RF-field inhomogeneity than gates composed only of shorter $\pi/2$ rotations. The lower extracted fidelity of Z is therefore consistent with the fact that it is not implemented as a virtual frame update in this experiment, but as the three-pulse sequence $R_x(\pi/2)R_y(\pi)R_x(-\pi/2)$, which includes both multiple RF pulses and a central full π rotation. Thus, the IRB results qualitatively support the pulse-composition-aware model used in the simulation, while the reported values should still be interpreted as standard IRB estimates rather than exact isolated gate fidelities.



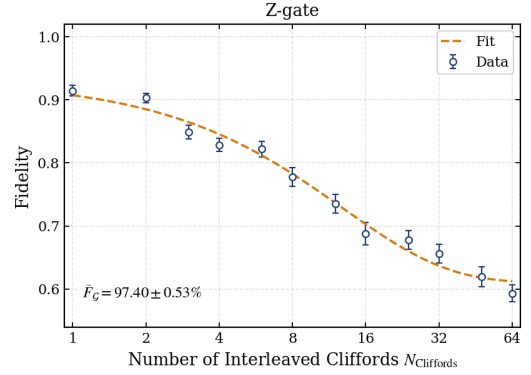
(a) X gate



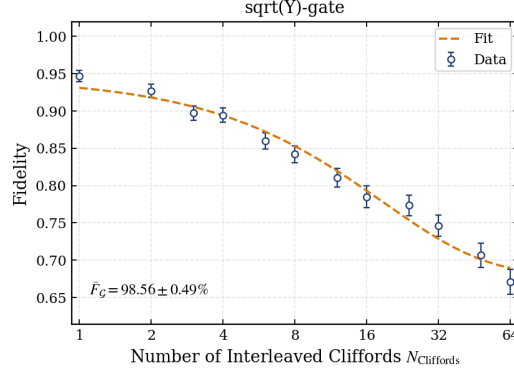
(b) H gate



(c) S gate



(d) Z gate



(e) $\sqrt{Y}$ gate

Figure 7: Interleaved RB decay curves derived from NMR experimental runs. Each panel shows the specific gate's probability-like interleaved decay signal relative to the Standard RB reference baseline.

4.4 Deterministic Benchmarking (DB) and Physical Limitations

The Deterministic Benchmarking (DB) protocol was executed to extract the underlying physical error parameters: the longitudinal relaxation time (T_1), the transverse relaxation time (T_2), and the coherent over-rotation ($\delta\theta$) and phase ($\delta\phi$) errors. Four specific sequence families ($\{\text{Free}; |1\rangle\}$, $\{XX; |+\rangle\}$, $\{YY; |+\rangle\}$, and $\{X\bar{X}; |+\rangle\}$), along with additional validation runs for $\{Y\bar{Y}; |+\rangle\}$ and $\{\bar{Y}Y; |+\rangle\}$, were applied up to a sequence length of $n_{\text{max}} = 130$, corresponding to an evolution time of $t = 2nt_g$. To ensure statistical reliability and generate appropriate error bars, each unique pulse sequence was independently executed 5 times.

Furthermore, to estimate the target fidelity for states like $|+\rangle$, we calculate $F(n) = \frac{1}{2}(1 + \text{Re}(S_n)/\text{Re}(S_{\text{ref}}))$; retaining the signed real signal instead of its magnitude is critical, as resulting fidelities below $1/2$ correctly capture the coherent oscillations when the final state becomes anti-aligned with the target.

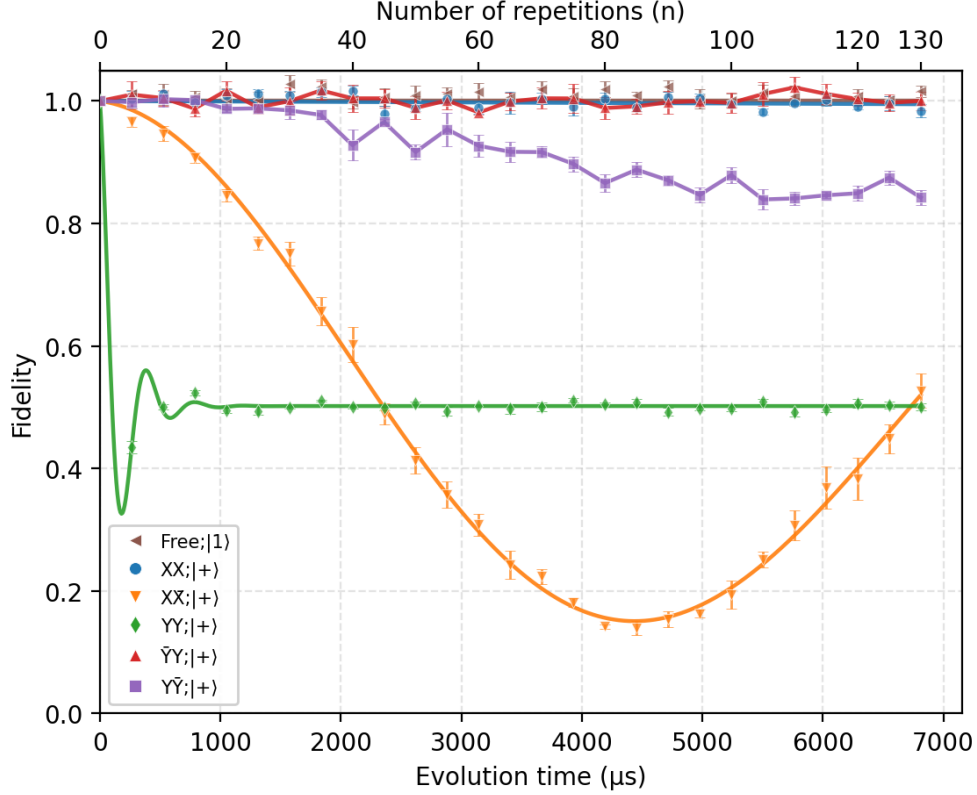


Figure 8: Deterministic Benchmarking fits from NMR experimental sequences. The plot illustrates the limited temporal extent constrained by safe RF pulsing bounds.

As shown in Figure 8, the measured probability-like signal for the $\{\text{Free}; |1\rangle\}$, $\{XX; |+\rangle\}$, and $\{YY; |+\rangle\}$ families remains near a fidelity of 1 throughout the entire $n_{\text{max}} = 130$ repetitions. The parameters extracted from the short-time empirical fits of these decay curves were:

- $T_1 \approx 2.27$ ms (from $\{\text{Free}; |1\rangle\}$)
- $T_2 \approx 1.29$ s (from $\{XX; |+\rangle\}$)
- $\delta\theta \approx 23.39^\circ$ (from $\{YY; |+\rangle\}$)
- $\delta\phi \approx 1.04^\circ$ (from $\{X\bar{X}; |+\rangle\}$)

However, these extracted parameters violate the fundamental physical bound $T_2 \leq 2T_1$, indicating that the empirical fits do not accurately reflect true open-system dynamics. This discrepancy arises due to stringent experimental duty-cycle constraints on the spectrometer. The true T_1 of the deuterated water sample is on the order of several seconds (specifically, 8.81 s as measured via inversion recovery in Section 4.3). We believe that had we been able to run the experiment in a longer timescale (of the order of seconds), then we would have been able to observe the decay.

Because the DB schedules are restricted to $n_{\text{max}} = 130$ pulses ($t \sim$ milliseconds) to prevent hardware stress and duty-cycle-induced heating from dense RF pulsing, the measurement window is orders of magnitude too short to capture the true exponential relaxation scales. Consequently, while these phenomenological DB fits summarize the short-time behavior of the current campaign, they cannot yet be used to generate a

full, physically validated Lindblad prediction for long-sequence NMR dynamics. Extending the DB protocol to the requisite multi-second regime remains fundamentally limited by the hardware’s safe RF continuous-wave capacity.

5 Conclusion

We developed and deployed an automated simulation-to-NMR pipeline for single-qubit benchmarking on a benchtop deuterated-water NMR system. The pipeline connects Clifford sequence generation, numerical simulation, R_x/R_y pulse compilation, and Prospa-compatible experiment generation. Standard RB yielded $p = 0.97614 \pm 0.00447$, corresponding to an average Clifford fidelity of $\bar{F}_C = 0.98807 \pm 0.00223$ and an average Clifford error rate of approximately 1.19%.

IRB provided gate-dependent fidelity estimates for S , H , X , $\sqrt{Y}$, and Z . The observed ordering is not explained by pulse count alone, but by the physical pulse decompositions. Gates involving full π rotations, especially the synthesized $Z = R_x(\pi/2)R_y(\pi)R_x(-\pi/2)$, are more sensitive to RF amplitude and calibration errors than gates composed only of shorter $\pi/2$ rotations. Thus, the IRB results support a pulse-realization-based interpretation of gate fidelity on this NMR platform.

DB exposed the main hardware limitation of the apparatus. Although inversion recovery measured $T_1 = 8.81$ s, safe DB pulse trains were limited to millisecond-scale durations by RF duty-cycle constraints and signal degradation. As a result, the DB fits summarize short-time behavior but cannot be interpreted as a fully validated extraction of long-time Lindblad parameters. Future work should use more frequent local FID references, improve π -pulse and axis calibration, explore virtual Z implementation, and modify DB schedules to access longer relaxation times without dense continuous RF driving.

6 References

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